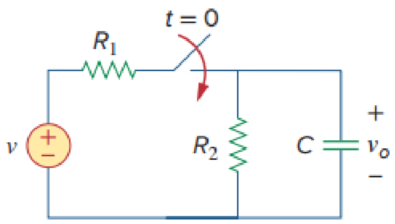
Chapter 7, Problem 41P

Problem

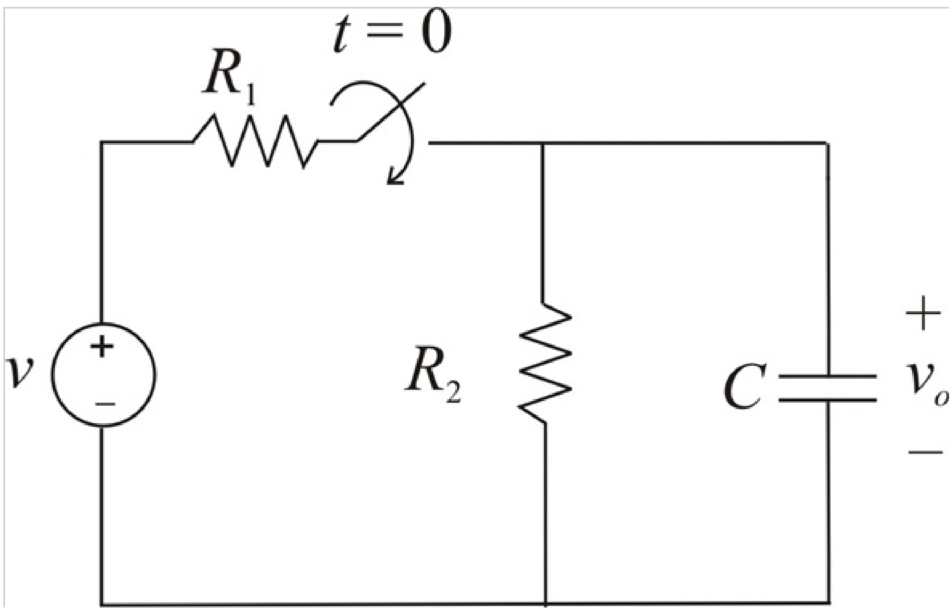
Using Fig. 7.108, design a problem to help other students better understand the step response of an *RC* circuit.



Step-by-step solution

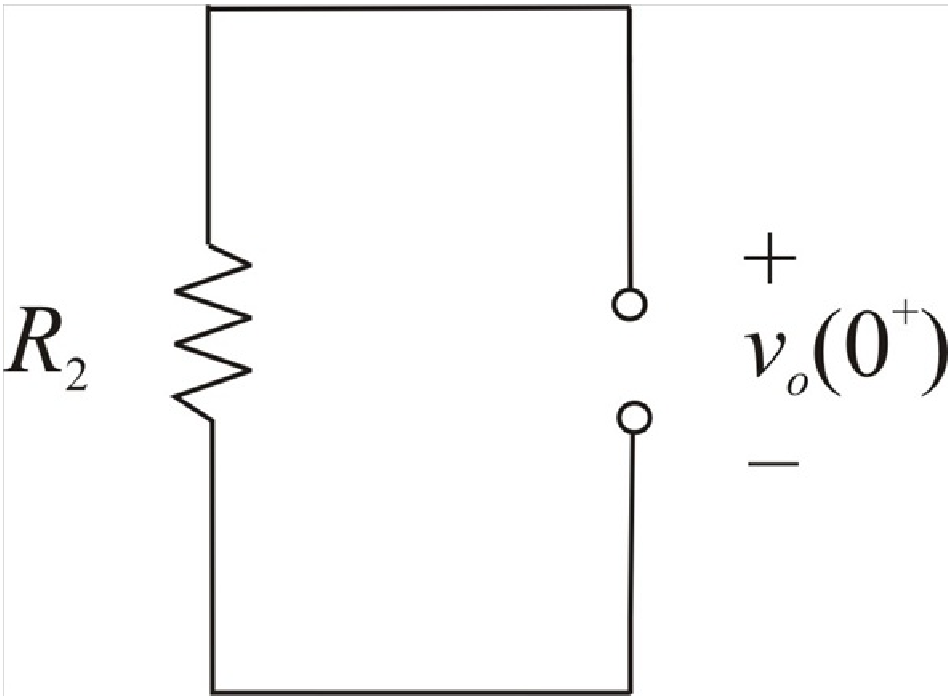
Step 1 of 7

Given circuit diagram



Step 2 of 7

For $t<0$ switch is opened and the capacitor acts live an open circuit for dc, hence the given circuit is as below



Step 3 of 7

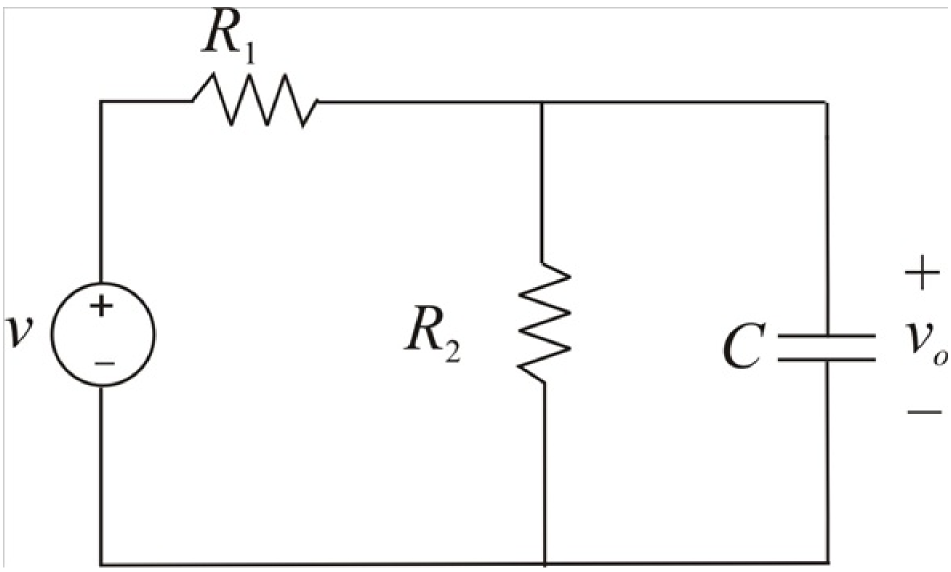
$v_C(0^-) = v_o(0^-) = 0\text{V}$

In capacitor voltage can't change instantaneously

$v(0) = v_C(0^+) = v_C(0^-) = 0\text{V}$

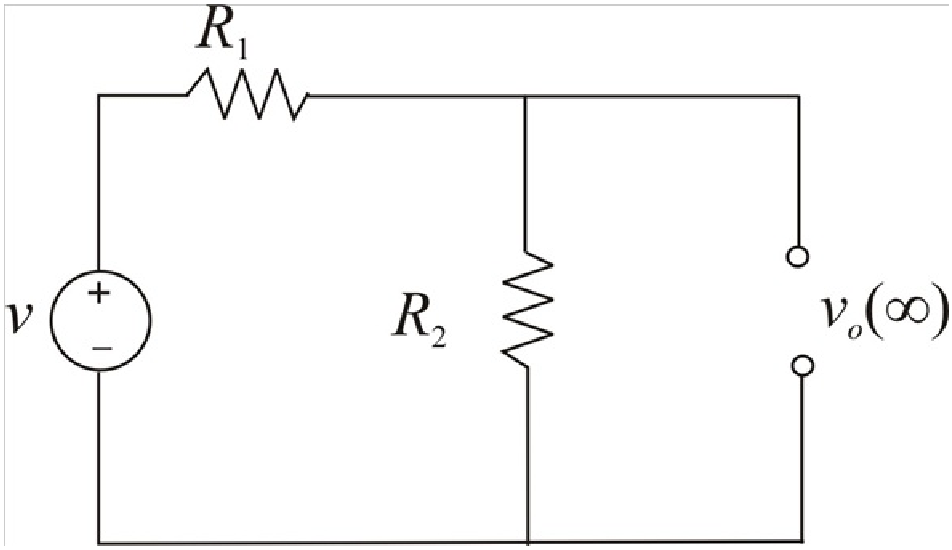
Step 4 of 7

For $t>0$ switch is closed hence the given circuit is as below



Step 5 of 7

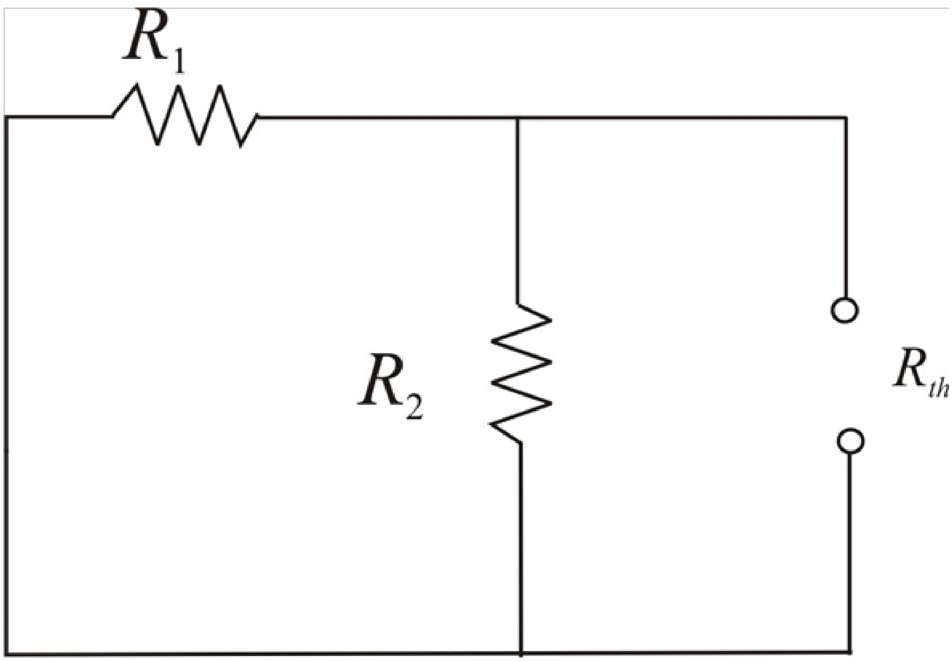
If the switch is closed for a long time then the circuit is reached to steady state and the capacitor acts as open circuit again.



Step 6 of 7

$$v_o(\infty) = v \times \left(\frac{R_2}{R_1 + R_2} \right)$$

R_{th} can be obtained by short circuiting the voltage source



Step 7 of 7

$$R_{th} = \frac{R_1 R_2}{R_1 + R_2}$$

Time constant

$$\tau = R_{th} C$$

$$\tau = \frac{R_1 R_2}{R_1 + R_2} C$$

Capacitor voltage

$$v_o(t) = v_o(\infty) + \left[v_o(0) - v_o(\infty) \right] e^{\frac{-t}{\tau}}$$

$$v_o(t) = v \times \left(\frac{R_2}{R_1 + R_2} \right) + \left[0 - v \times \left(\frac{R_2}{R_1 + R_2} \right) \right] e^{\frac{-t}{\left(\frac{R_1 R_2 C}{R_1 + R_2} \right)}}$$

$$\boxed{v_o(t) = \frac{v R_2}{R_1 + R_2} \left(1 - e^{\frac{-t(R_1 + R_2)}{R_1 R_2 C}} \right) \mathrm{V}} \text{ for } t > 0$$